

AWI-ESM3 LR: estimated CMIP7 output per model year

TCo95 + FESOM 2.7 core3 + REcoM + LPJ-GUESS | scaled from the measured HR test year cli119 | 15 September 2026

Result

LR writes everything HR writes, plus the rules deactivated for HR, plus REcoM. That is **about 600 to 710 files per model year**, against 502 for HR, and **about 110 GB per year** if daily 3D biogeochemistry is left out, or **about 160 GB** with it. The grids are much smaller than HR, so the larger file count comes to between a quarter and a third of HR's 470 GB.

600–710	502	≈ 110 GB	≈ 160 GB	11–16 TB
files per year	HR files per year	per year, monthly REcoM	with daily 3D REcoM	per 100 years

What LR writes

part	files per year	GB per year
Everything HR writes, scaled to LR grids (table below)	502	41
Reactivated: 6-hourly atmosphere on model levels (ta, ua, va, hus, zg)	5	30–50
Reactivated: 3-hourly atmosphere on 6 pressure levels (ta, ua, va, hus, wap)	5	8
Reactivated: 6-hourly atmosphere on 7 pressure levels (ta, ua, va, hus, zg), fixed for LR	5	4
Reactivated: ocean tendencies (6, yearly), msftmmpa (2, GM on in LR), daily hfx/hfy (2), sea-ice stress (2), soil moisture variants (3), sub-daily ps/ts (2), grass height (1)	18	< 3
REcoM, monthly ocean biogeochemistry as currently written	69–178	10–13
total without daily 3D biogeochemistry	604–713	≈ 100–120
REcoM daily 3D fields requested by CMIP7 (detoc, dissoc, phycalc, phypico, zmeso, zmicro, zooc)	(within 178)	+50

Not per year: 17 fixed fields once per run, 9 decadal files once per decade, one climatology.

Everything HR writes, scaled to LR

HR's cmorized volume was measured per grid on one full model year and divided by the ratio of cells, for 3D ocean fields also by the ratio of vertical levels. Coarse grids compress slightly worse per value and the fixed overhead per file does not shrink, so this part lies between 40 and 55 GB.

grid	HR	HR cells → LR cells	LR
native atmosphere and land	274 GB	TCO319 421 120 → TCO95 40 320 (÷10.4)	26.3 GB
regridded atmosphere	68 GB	1296×640 → 400×192 (÷10.8)	6.3 GB
FESOM nodes, 2D and 3D	110 GB	DARS 3 146 761 → core3 220 509, 57 → 48 levels	7.1 GB
FESOM elements, 2D and 3D	18 GB	6 229 020 → 426 403	1.1 GB

How the added parts were estimated

- **Reactivated rules.** 44 rules are commented out for HR; 11 of them are produced by another active rule already, which leaves 33 new files. Sizes of the sub-daily atmosphere sets come from HR's measured compressed bytes per stored value (about 1 byte on model levels, 1.7–2.7 bytes on pressure levels), applied to the LR grid. The model-level range reflects 91 or 137 vertical levels, which was not confirmed for LR.
- **REcoM, files.** The test run RECOM_concdriven_04 writes 69 biogeochemistry streams. The CMIP7 data request has 178 ocean biogeochemistry entries, 44 of them Southern Ocean subsets, and several entries derive from the same stream (surface values, integrals, sums over plankton groups). The real count is likely near 178.
- **REcoM, volume.** The 69 streams are 11 GB per year of raw output, all monthly. Cmoring keeps 82–94 % of monthly 3D ocean fields (measured on HR temp and salt), about 10 GB, plus a few GB for surface and regional entries. The seven daily 3D fields are about 7.6 GB each and need daily REcoM output.

Not covered

Compute time (HR's 33 node-hours per year is I/O-bound and does not scale reliably) and the length of each LR experiment; all numbers are per model year.

Sources: HR /scratch/a/a270092/pycmor_hr/cli119; LR mesh /work/ab0246/a270092/input/fesom2/core3; deactivated rules in awi-esm3-veg-hr-variables/;

LR atmosphere and REcoM /work/bb1469/a270092/runtime/awiesm3-develop-cc/ (cc_tco95_c, RECOM_concdriven_04)